



**CITIES SERVICE COMPANY
CITCO UNION TEXAS
PETROLEO DO BRASIL, LTDA.**

*duplicate
26.5 055*

**ACS-16 CONTRACT
WELL PROGNOSIS 1-APS-47
FOZ DO AMAZONAS BASIN
BRAZIL**

**EXPLORATION STAFF
DECEMBER 1980**

A. PROSPECT DATA SHEET

WELL: 1-APS-47

CLASSIFICATION: New Field Wildcat

RIG: Key Gibraltar - jackup

PROSPECT TYPE: Stratigraphic (carbonate buildup)

LOCATION: Line ES-90-29, Shotpoint 85
Approximately 625 Kms north of Belem

COORDINATES: Latitude: 2°28'3.3" N
Longitude: 49°02'26.5" W

WATER DEPTH: 40 meters (131 ft.)

OBJECTIVE: } To test the hydrocarbon potential
of Paleocene carbonate reservoirs

DEPTH TO OBJECTIVE: → 3.480 seconds (TWT)
5327 meters (17,478 ft.)

AREAL CLOSURE: 68.8 Km² (17,000 acres)

VERTICAL RELIEF: 225 meters (738 ft.)

PREDICTED TOTAL DEPTH: 5600 meters (18,374 ft.)

ESTIMATED COST: \$13,773,000

ESTIMATED DRILLING TIME: 145 days

PREDICTED SPUD DATE: January 2, 1981

DIVISION OF INTEREST:

Cities Service	37.5%
Union of Texas	25.0%
Longhorn	18.75%
Nicor	18.75%

1-APS-47

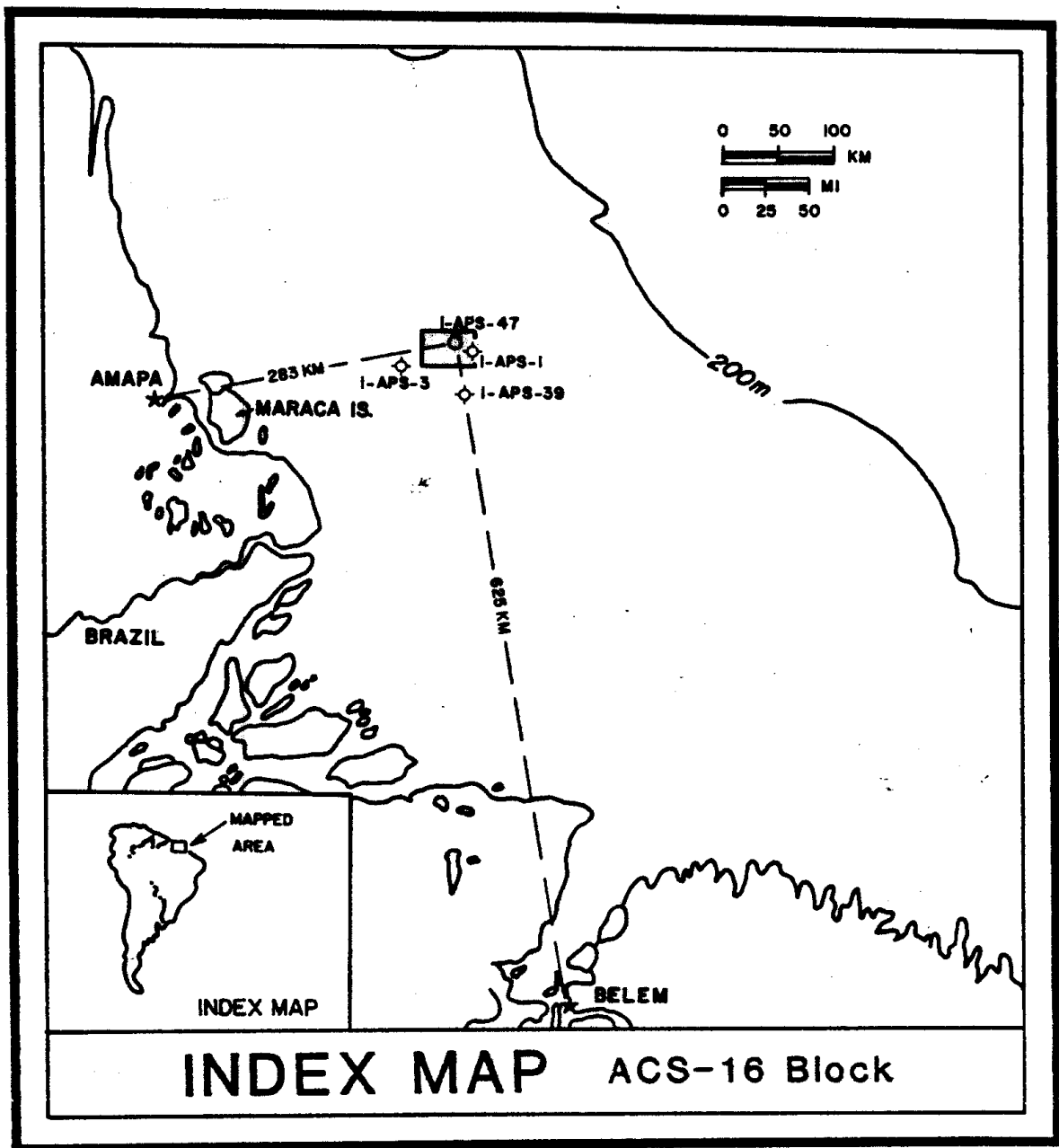
RECOMMENDATION TO DRILL

B. INTRODUCTION

The proposed location for the 1-APS-47 wildcat well lies approximately 625 km north of Belem, Para, in the northeastern corner of the ACS-16 block (Figure 1). Water depth in the vicinity is estimated to be 40 meters. Total depth for the well is projected at 5600 meters.

The prospect is intended to test the hydrocarbon potential of a Paleocene carbonate buildup indicated by seismic. The structure has apparent areal closure of 68.8 Km², and vertical relief of 225 meters. The top of the Paleocene is estimated to occur at 5327 meters.

FIGURE 1



C. REGIONAL SETTING

1. Location

The Foz do Amazonas Basin is located along the equatorial continental margin of Brazil, off the mouth of the Amazon River. Tectonic boundaries consist of the Guiana Shield to the west and the Brazilian shield to the south. To the southwest it is separated from the Paleozoic Amazon Basin by the Gurupa Arch.

2. Geologic History

Early in the Cretaceous, the present margin of Brazil evolved from a rift within the central Gondwana landmass. By Barremian time, substantial separation had occurred along the northern coast and a true taphrogenic margin was developed. For most of Barremian through Aptian time, continental clastics began to fill the developing rift basins. These deposits are recognized today along much of the northern coast of Brazil and the adjacent African margin.

An Albo-Aptian transitional facies from the Ghana coast probably reflects South Atlantic widening at this time. The advance of seas from the north may have been responsible for the transitional sediments of Albian age located in the Barreirinhas Basin. This equivalent sequence has yet to be encountered in the Foz do Amazonas Basin. By Cenomanian time a true marine environment was present off much of the Brazilian and African coasts. A minor regression apparently ensued and

was followed by a transgression in the Coniacian which endured, despite local interruptions, throughout the remainder of Cretaceous time. Permanent circulation between the North and South Atlantic was established in the Santonian.

In the Foz do Amazonas Basin only two formations are identified as Cretaceous in age. The Jacarezinho is a fault bounded fan conglomerate whose distribution is restricted by the grabens in the vicinity of Marajo Island. Its age, based upon stratigraphic considerations, is estimated to span all of Upper Cretaceous time.

The Limoeiro Formation is both laterally and vertically contiguous with the Jacarezinho. Its age, based on palynology, has been assigned to the Upper Cretaceous and Paleocene. It represents a finer and more mature clastic sediment whose character is entirely continental in the vicinity of the Marajo Basin.

The only indication of a possible marine equivalent to the type Limoeiro is encountered in the 1-PAS-1 and 1-PAS-5 wells at depths of 3200 meters and 3400 meters, respectively. Each well penetrated a substantial section of interbedded limestones, shales, and sandstones. We believe these sections verify a gradual basinward change of facies in the Paleocene and underlying strata. Thus, a continental environment is observed to grade to a near marine situation somewhere in the vicinity of the ACS-16 block. It's entirely conceivable that the contract area corresponded to a Cretaceous environment in which marine processes dominated.

The linear anomaly observed from the seismic record and discussed elsewhere easily lends itself to a carbonate shelf edge model. In view of the source potential of Cretaceous marine rocks, such an anomaly proves to be a target worthy of a deep test as proposed.

D. 1-APS-47 PROSPECT

1. Prospect Summary

The 1-APS-47 location lies in the northeastern corner of the ACS-16 block, 6.35 km north-northwest of the 1-APS-1 well. The objective is a Paleocene carbonate buildup on an ancient Cretaceous-Paleocene high energy shelf edge. Thermally mature Cretaceous marine shales are believed to be the primary potential source rock. Eocene shales draped over the reef structure should serve as a seal to the carbonate reservoir. This play concept remains untested in the vicinity of the ACS-16 area. Eocene and Oligocene carbonates are regarded as secondary objectives.

2. Geophysics

A total of 3205 kms of seismic was used in the interpretation of the ACS-16 block. Of this amount, 945 kms consist of reprocessed Petrobras data. The remaining 2260 kms were shot by Western Geophysical in 1979 and processed by Digicon. The data quality is considered good to fair down to a two-way time of 3.5 seconds. Time-structure and isochron maps were generated for several horizons and verify the presence of a linear trend which cuts the northeast corner of the ACS-16 block (Figure 3). This feature was interpreted as a high energy shelf edge carbonate of Paleocene and Cretaceous age. The proposed 1-APS-47 drilling location corresponds to a local 4-way closure representing a portion of the overall linear trend. This Paleocene reef structure has an apparent areal closure of 68.8 Km² (17,000 acres) and vertical relief of 225 meters.

The proposed location corresponds to shotpoint 85 of line ES 90-29 (Enclosure 2) and occurs near shotpoint 138 of line ES 90-4 (Enclosure 3).

Depths to the various horizons were derived by converting RMS velocities to interval velocities. Tertiary and Cretaceous horizons are estimated to occur at the following depths:

Lower Miocene Carbonates (yellow)	1.775 sec (TWT)	1863 m
Top Oligocene (red)	2.390	2832 m
Top Eocene (orange)	2.885	3891 m
Top Paleocene (pink)	3.480	5327 m
Approximate Top Cretaceous (lt. orange)	3.735	5864 m

X

E. DRILLING PARAMETERS

The jackup rig "Key Gibraltar" owned by Keydril Co., will drill 1-APS-47. Estimated date of spud is January 2, 1981.

1. Casing Program

(Casing points measured from Kelly bushing)

30" casing set at 400 ft. (122 m.)

20" casing set at 1,800 ft. (549 m.)

13 3/8" casing set at 7,500 ft. (2286 m.)

9 5/8" casing set at 14,000 ft. (4267 m.)

7" liner set at T.D. if required

5" liner at T. D. if required for testing

2. Mud Program

A pre-hydrated gel saltwater mud system will be used to a depth of 7,500 ft. (2286 m.)

A lignite, lignosulphonate, dispersed mud will be used from 7,500 ft. (2286 m.) to T. D.

3. Logging Program

Geoservice will continuously monitor the mud system and cuttings for presence of hydrocarbon shows, critical rig functions and downhole conditions.

Open and cased hole logs will be run by Schlumberger. The seismic reference survey will be the responsibility of C.G.G.

Coring, drillstem testing, and other evaluation services will be performed as required.

F. RESERVE ESTIMATES FOR 1-APS-47

BRAZIL - CONTRACT ACS-16

A. GEOLOGICAL PARAMETERS:

	<u>Low</u>	<u>Median</u>	<u>Mean</u>	<u>High</u>
<u>Geologic Input</u>				
Productive Acres	2,000	--	--	13,600
Net Pay, Feet	100	--	--	700
Recovery Factor, B/AF	100	--	200	300
<u>Monte Carlo Simulation Study (500 passes)</u>				
Gross Reserves, MMBbbls.	27	300	345	1,154

B. PHYSICAL PARAMETERS:

Field Size, MMBbbls.	300
Water Depth, Feet	50
Reservoir Depth, Feet	18,000
Distance to Shore, Miles	210
No. Producing Wells	32
Maximum Producing Rate	
Per Well, BOPD	5,000
For Field, BOPD	160,000
No. of Platforms (combination drlg./prod.)	1

BRAZIL -CONTRACT ACS-16

ECONOMIC SUMMARY
300 Million Barrel Field

	75% of Total Field Contractor's Interest*	
	<u>100%</u>	<u>Citco 37.5%</u>
Recoverable Reserves, MMBbls	75.3	28.3
Investment*		
Exploration Costs	14.8	5.6
Development Costs	908.5	340.7
Production Bonus	2.0	.75
Total	925.3	347.1
Remuneration to Contractor**		
Value at Market Price	5763.8	2161.4
Interest on Dev. Loan	418.2	156.8
Gross Revenue to Contractor	6182.0	2318.2
Brazil Income Tax	1544.1	579.0
U. S. Income Tax	1299.6	487.4
Net Cash Production	3323.5	1246.3
Present Worth Net Cash Production @ 15% Discount	21.8	8.2
Profitability Index, %/yr./yr.	25.9	25.9
Payout Time, Years	6.4	6.4
Payout Percent, %	352.6	352.6

* Inflation - Constant 1980 Dollars

** Price - 32 \$/Bbl.

G. CONCLUSION AND RECOMMENDATION

Prior to the drilling of the 1-APS-39, the objective of Cities and partners was the carbonates of Miocene through Eocene age. Since that time, industry has turned its focus to the older and, as yet, largely untested section which is believed to be present in the Foz do Amazonas Basin. The proposed location of the 1-APS-47 reflects Cities and partner's interest in the potential of Paleocene and Upper Cretaceous rocks which appear to be present in the ACS-16 block.

The trap size and potential reservoir section may contain a significant hydrocarbon accumulation and justifies an exploratory test as discussed here.

It is recommended that the 1-APS-47 wildcat be drilled as proposed.

APPENDIX

1 - APS - 47

1 - APS - 47 A

DRILLING OPERATIONS REPORT

AVERAGE MUD PROPERTIES

<u>PHASE</u>	<u>DEPTH (FT.)</u>	<u>DENSITY (ppg)</u>	<u>FUNNEL VISC. (sec.)</u>	<u>WATER LOSS (API)</u>	<u>REMARKS</u>
1	-	-	-	-	Sandbagging
2	436	7.9 - 8.3	100	-	Drive 30" csg.
3	1818	7.9 - 8.3	86-56	20 - 14	26" hole
4	7338	8.2 - 9.2	80-32	20 - 6.5	17 1/2" hole
5	14714	8.7 - 10.3	67-36	22 - 4.9	12 1/4" hole
6	14714	9.4 - 10.1	48-33	17.2 - 8.6	Working in 9 5/8" casing

CASING AND LOGGING

Casing was set as follows:

<u>Description</u>	<u>Depth from Rotary (ft.)</u>	<u>Shoe Test (ppg)</u>	<u>Remarks</u>
30", 1-1/2" wall, ALT	436	-	Driven-292' penetration 349 BPF. max.
20", 7/16" wall, L	1816	11.0	OBS hgr. at 200'-cem. to surface.
13-3/8", 72 ppf, N-80, Butt.	7319	15.1	OBS hgr. at 200'-cem. to hgr. DV tool at 5531'.
9-5/8", 47 ppf, P110, Butt.	14706	-	Cem. to 3873', DV tool at 10007'.

Logs were run as follows:

<u>Log</u>	<u>Depth from Rotary (ft.)</u>
ISF - Sonic	7323 to 1816
ISF - Sonic	14714 to 7319
Dipmeter	14714 to 7319

CEMENTING

Primary cementing, squeezes and plugs were as follows with class G cement:

DESCRIPTION	REMARKS
1. 169 sxs. mixed w/2% calcium chloride and 0.635 gal/sx Econolite at 11.5 ppg	468' - 421' to consolidate sand below 30" casing.
2. a. 1900 sxs mixed with 0.635 gal/ sx Econolite at 11.4ppg.	Lead cement - 20" casing.
b. 960 sxs mixed w/ 2% CaCl ₂ at 15.8 ppg.	Tail cement - 20" casing.
3. 107 sxs mixed w/ 2% CaCl ₂ at 15.8 ppg.	Squeeze 20" casing shoe, final pressure was 410 psi.
4. a. 1429 sxs mixed w/ 0.2% HR-4 and 0.75% CFR-2 at 15.8 ppg.	1st stage 13 3/8" casing.
b. 1846 sxs mixed w/ 0.5% HR-4 and 0.75% gal/sx Econolite at 11.4 ppg.	2nd stage 13 3/8" casing.
5. a. 450 sxs mixed w/ 0.15 gal/sx CFR-2L, 0.4% HR-12 and 0.1% CFR-1 at 15.2 ppg.	Side track plug 1. Set from 12,115 to 11,522 ft. Density to low.
b. 450 sxs mixed w/ 0.15 gal/sx CFR-2L, 0.4% HR-12 and 6% sand at 15.0 ppg.	Sidetrack plug 2. Set from 12,039 to 11650 ft. Density to low. Remaining rig stock blown over board.
c. 450 sxs mixed w/ 0.4% HR-4 at 16.1 ppg.	Sidetrack plug 3. Set from 12,039 to 11,428 ft. Plug softer than limestone formation.
d. 550 sxs mixed w/ 0.2% HR-12 and 0.1 gal/sx CFR-2L at 16.1 ppg.	Sidetrack plug 4. Set from 9901 to 9117 ft. Plug was contaminated with mud.
e. 478 sxs mixed w/ 0.4% HR-4, 0.29 gal/sx CFR-2L and .002 gal/sx NF-1 at 16.5 ppg.	Sidetrack plug 5. Set from 9783 to 9499 ft. Used spacer fluid and got a good plug.
6. a. 2000 sxs mixed with 0.4% Halad-9 0.6% CFR-2, 0.6% HR-12 at 15.8 ppg.	1st stage 9 5/8" cement, lost 370 bbls mud.
b. 1655 sxs mixed w/ 0.635 gal/ sx Econolite and 0.1% CFR-1 at 12.5 ppg.	2nd stage 9 5/8" cement, had full returns.

DESCRIPTION	REMARKS
7. a. 75 sxs mixed w/ 0.4% Halad-9, 0.2% gal/sx CFR-2L, 0.6% HR-12 at 15.8 ppg.	First P & A plug set from 14,615 to 14,405 feet.
b. 84 sxs mixed w/ 0.5% HR-4, 0.3% gal/sx CFR-2L at 15.8 ppg.	Second P & A plug set from 10,100 to 9864 feet.
c. 168 sxs mixed w/ 20% sand and 0.15% HR-12 at 16.0 ppg.	Third P & A plug set from 7600 to 7130 feet.
d. 107 sxs mixed w/ 20% sand and 0.2% HR-4 at 16.0 ppg.	Fourth P & A plug set from 6900 to 6600 feet.
e. 71 sxs mixed w/ 20% sand at 16.0 ppg.	Fifth P & A plug set from 5100 to 4900 feet.
f. 71 sxs mixed w/ 20% sand at 16.0 ppg.	Sixth P & A plug set from 4300 to 4100 feet.
g. 131 sxs mixed w/ 20% sand at 16.0 ppg.	Seventh P & A plug set from 3900 to 3700 feet.
h. 168 sxs mixed w/ 2% CaCl_2 at 16.0 ppg.	Eighth P & A plug set from 368 to 168 feet.

1 - APS - 47 and 1 - APS - 47 A

SUMMARY

The 1 APS 47 drilling operation began on December 27, 1980.
The location was:

Latitude: 02° 28' 05.019" North
Longitude: 49° 02' 23.993" West

in PSAD - 56 DATUM this was 46.6 meters from the proposed location.

Thirty inch casing was driven and the well was spudded on December 31, 1980. The 26" hole was drilled to 913 ft. At this depth the drill string became stuck. Fishing operations were conducted for several days with the drill pipe parting after attempting to back off. The well was plugged and abandoned with 118 ft of fish in the hole.

After plugging the 1 APS 47 well, the derrick was skidded back and the 1 APS 47 - A began at 0600 hours on Jan. 5, 1981.

Drilling had continued to a depth of 1257 ft on Jan. 13, 1981 when it was decided to jack the rig down and re-preload due to the barge settling. The string was pulled and the derrick was skidded back.

From Jan. 14 to Jan. 21 all operations were held up due to weather. On Jan. 15 Keydril informed the Citco company man on the rig, that for reasons of rig safety, Keydril had decided to move the rig. The rig was jacked down at 0930 hours on Jan. 21, 1981 and was removed from 1 APS 47 - A at 10:00 hours.

The Key Gibraltar was moved to the 1 APS 47 - B location at 10:00 hours on Jan. 21, 1981 and the well was drilled to a depth of 14,714 feet and abandoned after 151 days of operation on June 20, 1981.

DRILLING COST FOR 1-APS-47 and 1-APS-47-A

These two abortive wells were started on Dec. 27, 1980 and Jan. 5, 1981 respectively. They were ended on Jan. 5, 1981 and Jan. 21, 1981 respectively.

The 1 APS 47 took 9 days and cost approximately \$ 1,379.000* or \$ 153,222.00 per day. This well was abandoned after becoming stuck and having the drill pipe part during fishing operations at a depth of 913 ft.

The 1 APS 47 A took 16 days; the majority of this time was spent waiting on weather. The well was abandoned when KEYDRIL decided the rig was in danger due to some washing out around the spud cans. Total cost was approximately \$ 1,504.000.00* or \$ 94,000.00 per day.

The AFE's for these wells were revised according to the revision estimates on the following Detailed Well Estimates.

* Final cost will not be available until the AFE is closed.

1 - APS - 47 OPERATIONS REPORT

The Key Gibraltar arrived on location at 0600 hours on Dec. 28, 1980. It was spotted on location using the Mammoth Tide, the Supreme Tide, and the Defender. The legs were pinned and the rig was jacked up to a 5 ft air - gap and the preload was taken on. The rig location was determined to be:

02° 27' 50.70" North
by
49° 02' 30.30" West

This was approximately 100 feet south from the proposed location on a heading of 310°. Leg penetration was 22 ft on the port and starboard legs and 24 ft on the bow leg. The derrick was skidded out and rigged up to drill.

At this time most of the initial equipment and supplies were on the workboats that were at the location. This included all of the equipment that would be needed thru the second phase of the well, which included the setting of the 20" casing. Materials were off loaded and the driving of the 30" was begun at 1100 hours on Dec. 29. The 30" was driven to a total penetration of 201 ft below the mudline, with the butt weld sub 27 ft below the mudline. The 30" drive pipe was cut and the first bit and bottom hole assembly was picked up.

The hole was spudded at 0200 hrs on Dec. 31, 1980. Within 10 hours of spudding the well, mud losses reached 300 bbls per hour, so the pumps were switched over to sea water. The hole was drilled to 912 ft with sea water; pumping 30 bbls of high vis mud on each connection. At 912 ft the pipe became stuck while making a connection. The pipe was worked and various high viscosity muds were pumped around the pipe. At 0400 hrs on Jan. 3, 1981 the pipe parted. The string was pulled and 5 joints of drill pipe and 21 feet of the sixth joint was retrieved. The top of the fish was at 193 ft with 118 feet of pipe making up the fish. An overshot and five 6 1/2" drill collars were picked up and worked over the fish. The fish was jarred on with no success. A free point survey was run and the free point was found at 450 ft. Three string shots were run and on the third shot the pipe parted. The string was pulled and the overshot and fish were laid down. Fish recovered was 40 ft of drill pipe and 217 ft of heavy weight drill pipe.

Drill pipe was run open ended and a cement plug was set from 311 to 190 ft. The drill string was pulled and layed down. The diverter was nippedled down and the 30" casing was cut at 184 ft and pulled.

The rig was released from 1 APS 47 at 0600 hrs on January 5, 1981.

CHRONOLOGICAL EVENTS

Dec. 27, 1980
Day 1

Accepted rig at 0600 hours.

Dec. 28, 1980
Day 2

Arrived on location and varified same. Jacked up to 5 ft air gap, pre-loaded and held for 2 hours. Dumped pre-load and jacked up to 34 ft air gap. Skidded derrick and rigged up drill floor.

Preliminary location was
2° 27' 50.70" N
49° 02' 03.30" W

or approximately 100 ft south of the proposed location.

Dec. 29, 1980
Day 3

Rigged up floor to run 30" drive pipe. Picked up 5 joints and had 2 joints of free fall. Drove 30" to a total depth of 339 ft. Total penetration below mudline was 201 ft. Mud line hanger at 162 ft.

Dec. 30, 1980
Day 4

Cut and layed down 30" pipe. Nippled up hydril, diverter valves and bell nipple. Function tested 29 1/2" diverter and started to pick up BHA.

Dec. 31, 1980
Day 5

Drilled to 367 ft and cleaned hole. Drilled to 393 ft and cleaned hole. Drilled to 423 ft and lost 300 bbls of mud per hour. Switched to sea water. Drilled to 480 ft and dropped survey, 1/4°. Drilled to 672 ft with sea water. Flushed with mud on connections.

Jan. 1, 1981
Day 6

Drilled from 673 to 912 ft with sea water. Pumped 30 bbls high viscosity mud on each connection. Pipe stuck while making a connection.

Jan. 2, 1981
912 ft - 8.7ppg
Day 7

Worked stuck pipe with various weights from 25,000 to 350,000 lbs. Mixed and circulated high vis mud around while working pipe. Stuck at 886 ft.

Jan. 3, 1981
912 ft - 8.7ppg
Day 8

Pipe parted while working pipe. POOH and laid down 5 jts of drill pipe and 21 feet of the sixth joint. Fish left in hole was 118.50'. Picked up overshot, jars, and five 6 1/2" DC. Worked over fish and jarred on with no success. Ran Schlumberger to run free points. Found free point at 450 ft. Attempted to back off, no success. POOH and laid down overshot.

Jan. 4, 1981
Day 9

Recovered partial fish. Laid down 40 ft of drill pipe and 216 ft of heavy wright drill pipe. RIH and washed from 361 to 480 ft. Pipe stuck and worked free. Set cement plug with 450 sx class G cement plus 2% CaCl mixed at 15.8 ppg. Set plug from 308 to 187 ft. POOH and nipped down diverter. Cut 30" at 184 ft.

Jan. 5, 1981
Day 10

Laid down 30" casing. Broke and laid down 30" cutting and skidded derrick 8 ft. Rig released at 0600 hrs.

HELICOPTER FLIGHT SUMMARY

1 - APS - 47

DATE	FLIGHT	Nº PASSENGERS	WEIGHT (KG)	REMARKS
12.29.80	1	13 - 14	768 - 1109	2
	2	16 - 15	1171 - 1058	1,2
	3	15 - 15	1181 - 1075	2
12.30.80	4	11 - 12	997 - 1083	2
01.02.81	5	8 - 4	747 - 374	3
01.03.81	6	1 - 0	278 - 0	1

1. Citco Personnel.
2. Keydril Personnel.
3. Third Party Personnel.

BOAT VOYAGE - CARGO SUMMARY

1 - APS - 47

Load NBC	Manifest Number	Date	Boat	Value US\$	Weight (Long Tons)	Remarks
1	001	12 Dec. 80	Defender	282,076.16	66.25	A, B, C.
2	002	22 Dec. 80	Defender	24,914.14	.6	D, E
3	003	31 Dec. 80	Sup. Tide	313,369.92	229.6	A, B, D, E, F, G
4	004	03 Jan. 81	Defender	74,095.42	123.80	B, G
			TOTAL	694,455.64		

A: Drilling Equipment

B: Fuel

C: Navigational Equipment

D: Lube Oil

E: Diesel

F: Supplies & Spare Parts

G: Jet Fuel

H: Cement

MATERIALS	CONSUMED
1. 100% Cotton	100%
2. 100% Polyester	100%
3. 100% Nylon	100%
4. 100% Rayon	100%
5. 100% Silk	100%
6. 100% Wool	100%
7. 100% Linen	100%
8. 100% Leather	100%
9. 100% Rubber	100%
10. 100% Glass	100%
11. 100% Metal	100%
12. 100% Plastic	100%
13. 100% Paper	100%
14. 100% Wood	100%
15. 100% Stone	100%
16. 100% Concrete	100%
17. 100% Brick	100%
18. 100% Cement	100%
19. 100% Asphalt	100%
20. 100% Steel	100%
21. 100% Aluminum	100%
22. 100% Copper	100%
23. 100% Zinc	100%
24. 100% Iron	100%
25. 100% Lead	100%
26. 100% Tin	100%
27. 100% Silver	100%
28. 100% Gold	100%
29. 100% Platinum	100%
30. 100% Palladium	100%
31. 100% Nickel	100%
32. 100% Cobalt	100%
33. 100% Manganese	100%
34. 100% Chromium	100%
35. 100% Vanadium	100%
36. 100% Titanium	100%
37. 100% Bismuth	100%
38. 100% Antimony	100%
39. 100% Arsenic	100%
40. 100% Selenium	100%
41. 100% Tellurium	100%
42. 100% Iodine	100%
43. 100% Bromine	100%
44. 100% Chlorine	100%
45. 100% Fluorine	100%
46. 100% Oxygen	100%
47. 100% Nitrogen	100%
48. 100% Carbon	100%
49. 100% Hydrogen	100%
50. 100% Helium	100%
51. 100% Neon	100%
52. 100% Argon	100%
53. 100% Krypton	100%
54. 100% Xenon	100%
55. 100% Radon	100%
56. 100% Francium	100%
57. 100% Actinium	100%
58. 100% Thorium	100%
59. 100% Protactinium	100%
60. 100% Uranium	100%
61. 100% Neptunium	100%
62. 100% Plutonium	100%
63. 100% Americium	100%
64. 100% Curium	100%
65. 100% Berkelium	100%
66. 100% Californium	100%
67. 100% Einsteinium	100%
68. 100% Fermium	100%
69. 100% Mendelevium	100%
70. 100% Nobelium	100%
71. 100% Lawrencium	100%
72. 100% Rutherfordium	100%
73. 100% Dubnium	100%
74. 100% Seaborgium	100%
75. 100% Bohrium	100%
76. 100% Hassium	100%
77. 100% Meitnerium	100%
78. 100% Darmstadtium	100%
79. 100% Roentgenium	100%
80. 100% Copernicium	100%
81. 100% Nihonium	100%
82. 100% Flerovium	100%
83. 100% Moscovium	100%
84. 100% Livermorium	100%
85. 100% Tennessine	100%
86. 100% Oganesson	100%
87. 100% Ununseptium	100%
88. 100% Ununseptium	100%
89. 100% Ununseptium	100%
90. 100% Ununseptium	100%
91. 100% Ununseptium	100%
92. 100% Ununseptium	100%
93. 100% Ununseptium	100%
94. 100% Ununseptium	100%
95. 100% Ununseptium	100%
96. 100% Ununseptium	100%
97. 100% Ununseptium	100%
98. 100% Ununseptium	100%
99. 100% Ununseptium	100%
100. 100% Ununseptium	100%

BITS USED

1 APS 47

<u>Bit No.</u>	<u>Size</u>	<u>Type</u>	<u>S/N</u>
1	26	MILJ	31728

30" Drive Pipe 1 APS 47

Casing Driven 332.14 feet

Landing Joint Removed 31.43 feet

Casing in Hole 300.71 feet

KDB to Top of Csg 38.29 feet

Casing Depth 339.00 feet

30" Cut 186.00 feet

KDB to Top of Csg 38.29 feet

Csg Retrieved 147.71 feet

Casing Abandoned in Well and

Chargeable to 1 APS 47 153.00 feet

Mud and Cement Usage 1-APS-47

Bentonite		2560 Sacks
Caustic Soda		61 Sacks
Q Broxin		2 Sacks
Soda Ash		1 Sack
Cellosize		18 Sacks
Lime (Cal Bahia)		13 Sacks
CMC H 100		19 Sacks
Class G Cement		450 Sacks
Calcium Chloride		9 Sacks

(Taken from mud reports and daily drilling reports)

DIESEL SUMMARY

ON HAND

Key Gibraltar	238,475.8 liters
Mammoth Tide	243,427.0 liters
Supreme Tide	276,688.6 liters
Maersk Topper	266,192.0 liters
Gem Tide	207,057.5 liters
Johanistrum	<u>82,913.9 liters</u>
	1,314,754.8 liters
Purchased	<u>80,000 liters</u>
TOTAL	1,394.754.8 liters.

DETAILED WELL ESTIMATE

1 APS 47

DESCRIPTION	LAST EST.	REVISION Nº 1
TANGIBLES		US \$1000
Casing & associated equip. (size & qty.)		
30" 500 ft	254	200
20" 2000 ft	168	
13 3/8" 7500 ft	216	
9 5/8" 16000 ft	476	
7" 5400 ft	88	
5" 3500 ft	78	
3 1/2" 20000 ft	234	
Wellhead equipment	118	
Surface equipment		
Compressors		
Misc. tangible equipment	140	
TOTAL TANGIBLE EQUIPMENT	1772	200
INTANGIBLES		
Company labor		
Misc. contract labor	174	11
Mobilization		
Drilling rig	4350	270
Rig fuel	365	23
Oil, grease, pipe dope & other lubricants	70	1
Drilg. mud, diesel & chemicals	911	57
Mud logging unit & services	240	15
Cement & services	174	-
Cmt. squeeze jobs, plugging back, plug. well	45	45
Drill stem testing	175	11
Electric logs	331	20
Acidizing & fracing	50	
Perforating service		
Bits, reamers, diamond coring & analysis	180	6
Drilling & fishing tools	250	16
Rental Misc. Equipment		
Location Cost	250	250
Marine transportation	2550	309
Air transportation	1160	64
Land transportation	150	10
Warehouse expense	172	11
Diving service		
Communications	50	3
Insurance	204	13
Custom duties & misc. governmental fees	36	36
Misc. intangible material	57	4
Misc. expense	57	4
TOTAL INTANGIBLE COSTS	12001	1179
Total estimated cost - 100% U.S. Dls.	13773	1379
Total estimated C.S. - % U.S. Dls.		

DRILLING OPERATIONS

REPORT

After plugging and abandoning the 1 APS 47 and all the preparation work for the 1 APS 47-A had been completed on the rig floor, the derrick was skidded back and the 1 APS 47-A began at 0600 hours on January 5, 1981.

A string of 30" casing was driven to a depth of 474 ft. The 30" was cut and dressed and the 29-1/2" diverter was installed. A 26" bit and bottom hole assembly was picked up and drilling was started at 474 ft. Drilling was continued from 474 to 489 ft when total returns were lost. The drilling was continued to 450 ft with 20% returns.

A cement plug was set at 496 ft with 250 sxs of class G cement. Soft cement was found from 408 to 418 ft. The cement was drilled out and drilling continued to a depth of 585 ft with a mud weight of 8.4 mgg. this included 30% diesel in the mud. The hole was circulated and the string was pulled to the shoe, with the string hanging up at 541 ft. The barge was leveled by lowering the bow 20-3/4 inches.

The string was pulled for a bit change and drilling was continued from 585 to 1257 ft with thirty minute circulations on each connection. Drilling had continued to a depth of 1257 ft on Jan. 13 when it was decided to jack the rig down and re-preload due to the barge settling. The string was pulled and laid down and on January 13, 1981 the derrick was skidded back.

From January 14 to January 21 all operations were held up due to waiting on weather. The rig was put on Force Majeure on January 13 at 1200 hours. On January 15 Key Drill informed the Citco company man on the rig, that for reasons of rig safety it was Key Drill's decision to move the rig rather than re-preload. It should be noted that during this time Key Drill considered it safe to jack up and recover the 29-1/2" diverter but that they declaired it was not safe enough to plug the well and cut the 30" casing.

The rig was jacked down at 0930 hours on January 21, 1981 and was removed from 1 APS 47-A at 1000 hours.

CHRONOLOGICAL EVENTS

<u>Jan. 05, 1981</u> <u>Day 1</u>	Accepted the rig at 0600 hrs. Ran and drove 8 joints of 30" casing.
<u>Jan. 06, 1981</u> <u>Day 2</u>	Picked up joint N° 9 and drove an additional 10 feet. Drove 9 joints of 30" casing to a total depth of 416 feet, with 276 feet of penetration below the mudline. Hung off on the 30" and prepared to wash out.
<u>Jan. 07, 1981</u> <u>Day 3</u>	RIH with bit and washed and drilled to 430 ft. Lost 90% returns at 423 ft. Stuck pipe at 423 ft and was able to work free. Regained full returns while working pipe. Survey at 407 ft was 1-3/4°. Drove pipe to 446 ft. Washed to 446 ft and then drove the 30" to 476ft. Rigged down hammer and secured drive pipe.
<u>Jan. 08, 1981</u> <u>Day 4</u>	Out 30" casing and nipped up the diverter.
<u>Jan. 09, 1981</u> <u>Day 5</u>	Attempted to test diverter, but would not test. Repaired leaking bell nipple. Attempted to test again, but it still leaked. Replaced Hydril rubber.
<u>Jan. 10, 1981</u> <u>Day 6</u>	Finished working on diverter and tested to 250 psi. Drilled from 472 - 489 ft. Lost returns at 489 ft and and drilled to 499 ft with 20% returns. Surveyed 1° and pulled out of hole. RIH open ended to 492 ft and set a cement plug at 492 ft with 250 sxs of class G cement with 2% CaCl and .635 gal/sx Econolite at 11.4 ppg. POOH, made up BHA and ran in hole to 285 ft and displaced to 407 ft with 8.4 ppg mud (30% diesel). Washed and drilled soft cement.
<u>Jan. 11, 1981</u> <u>Day 7</u>	Drilled from 499 to 584 ft with frequent circulations. Pulled to shoe and hung up at 541 ft. Had to level barge by 20-3/4" on the bow leg due to settling. Drilled to 738 ft.
<u>Jan. 12, 1981</u> <u>Day 8</u>	Drilled to 1050 ft and surveyed 3/4°. Drilled to 1122 ft.
<u>Jan. 13, 1981</u> <u>Day 9</u>	Drilled to 1256 ft. Pooh and laid down drill pipe. Prepared to skid derrick back to re-preload.
<u>Jan. 14, 1981</u> <u>Day 10</u>	Waiting on Weather

Jan. 15, 1981
Day 11

Waiting on Weather. Maintaining a minimum air gap. Keydril informed Citco that for reasons of rig safety they had decided to move the rig rather than re-preload.

Declared Force Majeure as of January 13, 1981 at 1200 hrs.

Jan. 16, 1981
Day 12

Waiting on Weather

Jan. 17, 1981
Day 13

Waiting on Weather

Jan. 18, 1981
Day 14

Waiting on Weather

Jan. 19, 1981
Day 15

Waiting on Weather

Jan. 20, 1981
Day 16

Waiting on Weather

Jan. 21, 1981
Day 17

Tied on to Supreme Tide and Mammoth Tide for rig move. Rig released from 1-APS-47-A at 1000 hrs.

HELICOPTER FLIGHT SUMMARY

1 - APS - 47 A

DATE	FLIGHT	Nº PASSENGERS	WEIGHT (KG)	REMARKS
01.06.81	1	0 - 0	101 - 20	2
	2	2 - 4	204 - 420	2,3
01.08.81	3	0 - 2	420 - 200	2
01.09.81	4	2 - 0	268 - 0	1,2
01.12.81	5	18 - 15	1181 - 1181	2,3
01.12.81	6	8 - 11	577 - 692	2,3
01.13.81	7	9 - 4	1014 - 384	2,3
01.14.81	8	2 - 1	180 - 90	2
01.15.81	9	0 - 10	0 - 930	1,2,3
01.16.81	10	11 - 13	857 - 1014	1,2,3
	11	12 - 10	911 - 708	2,3
01.17.81	12	7 - 4	705 - 627	2,3
01.20.81	13	2 - 2	229 - 210	1,3

1 - Citco Personnel

2 - Keydril Personnel

3 - Third Party Personnel.

BOAT VOYAGE - CARGO SUMMARY

1 - APS - 47 A

LOAD No.	MANIFEST NUMBER	DATE	BOAT	VALUE US\$	WEIGHT (LONG TONS)	REMARKS
01	005	06 Jan 81	Sup. Tide	81,065.60	163.46	b,e,h
02	006	12 Jan 81	Mam. Tide	77,280.00	355.00	b,e,f
03	007	19 Jan 81	Jahannstrum	141,564.37	166.70	a,b,e,f,g
			TOTAL	299,909.97		

a. Drilling Equipment

b. Fuel

c. Navigational Equipment

d. Lube Oil

e. Diesel

f. Supplies & Spare Parts

g. Jet Fuel

h. Cement

MUD AND CEMENT USAGE 1-APS-47 A

Bentonite	610 Sacks
Soda Ash	5 Sacks
Caustic Soda	71 Sacks
Lime (Cal Bahia).....	4 Sacks
OMC H 100	2 Sacks
Spersene	36 Sacks
Cellosize	9 Sacks
Surflo W 300	4 Pails
DD Detergent	1 Drum
Aluminium Stearate	4 Sacks
Class 'G' Cement	250 Sacks
Calcium Chloride	5 Sacks
Econolite	3 Drums
Diesel for Mud	82,793 Gallons

Taken from Mud Report and Daily Drilling Reports.

MATERIALS CONSUMED

<u>Bit N°</u>	<u>Size</u>	<u>Type</u>	<u>S/N</u>
1	26	MILJ	31729

30" DRIVE PIPE

Casing Driven	476.88 feet
Landing Joint Removed	37.82 feet
Casing in Hole	439.06 feet
Csg left on location and chargeable to 1-APS-47 A.	439.06 feet

1 APS 47 A

DIESEL SUMMARY

Purchased

537,701 liters

DETAILED WELL ESTIMATE

1 APS 47 A

DESCRIPTION	LAST EST.	REVISION Nº 1
TANGIBLES		US \$1000
Casing & associated equip. (size & qty.)		
30" 500 ft		254
20" 2000 ft		
13 3/8" 7500 ft		
9 5/8" 16000 ft		
7" 5400 ft		
5" 3500 ft		
3 1/2" 20000 ft		
Wellhead equipment		5
Surface equipment		
Compressors		
Misc. tangible equipment		
TOTAL TANGIBLE EQUIPMENT		259
INTANGIBLES		
Company labor		
Misc. contract labor	174	18
Mobilization		
Drilling rig	4350	461
Rig fuel	365	39
Oil, grease, pipe dope & other lubricants	70	7
Drig. mud, diesel & chemicals	911	97
Mud logging unit & services	240	25
Cement & services	174	18
Cmt. squeeze jobs, plugging back, plug. well	45	5
Drill stem testing	175	19
Electric logs	331	35
Acidizing & fracing	50	5
Perforating service		
Bits, reamers, diamond coring & analysis	180	19
Drilling & fishing tools	250	26
Rental Misc. Equipment		
Location Cost		
Marine transportation	2550	270
Air transportation	1160	123
Land transportation	150	16
Warehouse expense	172	18
Diving service		
Communications	50	5
Insurance	240	22
Custom duties & misc. governmental fees	36	4
Misc. intangible material	57	6
Misc. expense	57	7
TOTAL INTANGIBLE COSTS	11751	1245
Total estimated cost - 100% U.S. Dls	13523	1504
Total estimated C.S. - % U.S. Dls		